

Concretely, let A_{ik} be the cell accuracy for technique $i \in \{1, \dots, I\}$ and paraphrase $k \in \{1, \dots, K\}$ ($I = 3$; $K = 4$ on GSM8K, $K = 2$ on HumanEval) at the reference decoding cell. We fit the balanced two-way variance-components model $A_{ik} = \mu + \alpha_i + \beta_k + \varepsilon_{ik}$, with $\alpha_i \sim (0, \sigma_{\text{alg}}^2)$, $\beta_k \sim (0, \sigma_{\text{lex}}^2)$, and $\varepsilon_{ik} \sim (0, \sigma_{\text{res}}^2)$, mutually independent, and estimate the variance components from the ANOVA mean squares by the method of moments:

$$\hat{\sigma}_{\text{alg}}^2 = \max\left(0, \frac{MS_{\text{tech}} - MS_{\text{res}}}{K}\right), \quad \hat{\sigma}_{\text{lex}}^2 = \max\left(0, \frac{MS_{\text{lex}} - MS_{\text{res}}}{I}\right). \quad (1)$$

Non-negativity truncation is what produces the degenerate AER values reported in Table ???. Confidence intervals on AER come from an item-level bootstrap rather than a closed-form interval because the truncated estimator has no standard asymptotic distribution at these small cell counts ($I = 3$, $K \leq 4$).

Because this comparison sets self-consistency at its required $T = 0.7$ against greedy ($T = 0$) baselines, the matched-temperature comparison—self-consistency against single-sample few-shot CoT, both at $T = 0.7$ —is reported separately in Table ?? and Fig. ??a (Section ??).